

Coulomb response regularity and the remaining three-label term

HOCF–RIESZ, Round 026 18 September 2026

Status. THM050 and THM051 have passed both complete fresh audit axes and root comparison. The original critical source assertion THM046 remains open.

Use the actual iid-Haar-prepared gradient particles on the unit four-torus,

$$dX_i = \frac{1}{N} \sum_{j \neq i} K(X_i - X_j) dt + \sqrt{2\nu} dW_i, \quad K = -\nabla g, \quad \hat{g}(m) = |m|^{-2} \ (m \neq 0), \quad \hat{g}(0) = 0.$$

Put $c = 4\pi^2$, $e_k(x) = e^{2\pi i k \cdot x}$, $\ell_k = c|k|^2$ and $F_k(x) = N^{-1} \sum_i e_k(x_i)$. Superscript ϵ means the actual dynamics with $K_\epsilon = p_\epsilon * K$. All singular passages in the first part take place at fixed finite $N \geq 2$ and $\nu > 0$.

Exact response costs. Let P_t denote the actual configuration semigroup, $a = c + \nu \ell_k$ and $A = e^{-aT}$. Conditional orthogonality under the original initial Haar law gives

$$\begin{aligned} D_N &:= N \mathbb{E} |F_k(X_T) - A F_k(X_0)|^2 = C_N + J_N, \\ C_N &:= N \|P_T F_k - A F_k\|_{L^2(dx^N)}^2, \\ J_N &:= N \mathbb{E} |F_k(X_T) - P_T F_k(X_0)|^2. \end{aligned}$$

These are nonnegative finite numbers. For $u_s^\epsilon = P_{T-s}^\epsilon F_k$, the actual smooth backward martingale has Brownian coefficients $\sqrt{2\nu} \nabla_i u_s^\epsilon$. Its conjugate bracket is $2\nu \sum_i |\nabla_i u_s^\epsilon|^2 ds$, and its cross bracket with the empirical forward martingale is

$$\frac{2\nu}{N} \sum_i \nabla_i u_s^\epsilon(X_s^\epsilon) \cdot \overline{\nabla e_k(X_i^\epsilon(s))} ds.$$

The frozen backward weight contributes $e^{-a(T-s)}$ when that weighted one-body martingale is used. It is not the cutoff-linearized response eigenvalue. Coupled bounded endpoint convergence proves

$$J_N = \lim_{\epsilon \downarrow 0} 2\nu N \int_0^T \mathbb{E} \sum_i |\nabla_i P_{T-s}^\epsilon F_k(X_s^\epsilon)|^2 ds.$$

This is a limit of costs, with no assertion about limiting singular gradients. Thus $D_N \rightarrow 0$ if and only if both $C_N, J_N \rightarrow 0$. At $k = 0$ the expressly artificial comparison $a = c$ gives $J_N = 0$ and $C_N = D_N = N(1 - e^{-cT})^2$; the physical constant-preserving response is unchanged.

A fixed-particle angular obstruction. Write $x_1 = z + r\theta/2$, $x_2 = z - r\theta/2$, $q = (z, x_3, \dots, x_N)$ and $H_k(q) = 2e_k(z)/N + N^{-1} \sum_{j \geq 3} e_k(x_j)$. Let Q be a compact positive-volume slow-coordinate set with separated neighborhoods. For some $C, t_0 > 0$, depending on these fixed data, every $0 < t \leq t_0$ satisfies

$$\limsup_{r \downarrow 0} \sup_{q \in Q, \theta \in S^3} \left| P_t F_k(q, r\theta) - H_k(q) + \frac{4\pi^2}{N^{3/2}} e_k(z) (k \cdot \theta)^2 \sqrt{t} \right| \leq Ct^{3/4}.$$

The relative drift is $4y/(N|y|^4) + O(y)$ and its noise covariance is $4\nu I$. Stopped radial moments, inverse-square occupation and angular Itô calculus give the displayed expansion while retaining every other particle. It needs no directional entrance limit.

For $f = \Re F_{(1,0,0,0)}$, the nonconstant angular profile persists on a positive-volume Q . Its full fourth-gradient cost contains a positive multiple of $t^2 \int_0^{r_t} dr/r$. Therefore, for every fixed N, ν , at every sufficiently small positive time,

$$P_t f \notin W^{1,4}(dx^N), \quad \liminf_{\epsilon \downarrow 0} \int |\nabla P_t^\epsilon f|^4 dx^N = \infty.$$

The second assertion follows on fixed annuli before shrinking their inner radius. This gives no $W^{1,2}$ conclusion and no interval uniform in N .

Actual pair estimates and the two-label cancellation. Fix $k \neq 0$. Write $Z_m(t) = F_m(X_t)$ and

$$g_r = p_r * g, \quad q_r = c \int_0^r p_u du, \quad a_r(m) = |m|^{-2} e^{-cr|m|^2} \quad (m \neq 0), \quad a_r(0) = 0.$$

The exact positive splitting $g = g_r + q_r - cr$, the actual energy sign $\mathbb{E} H_N(X_t) \leq 0$ and smooth self subtraction give, uniformly in all finite times and positive diffusivities,

$$\frac{N}{N-1} \mathbb{E} \sum_{m \neq 0} a_r(m) |Z_m(t)|^2 + \mathbb{E} q_r(X_1 - X_2) \leq \frac{g_r(0)}{N-1} + cr \leq C((Nr)^{-1} + r).$$

The first inequality holds for $0 < r \leq 1$ with its exact coefficients. The central heat contribution at $r = R^2$ yields

$$\mathbb{P}\{\text{dist}(X_1(t), X_2(t)) \leq R\} \leq C(R^4 + N^{-1}) \quad (0 < R \leq R_0 < 1/8).$$

This is a deterministic-time bound. Let

$$\begin{aligned} j_k(x, y) &= K(x - y) \cdot (\nabla e_k(x) - \nabla e_k(y)) + c(e_k(x) + e_k(y)), \\ j_{k,r}(x, y) &= K_r(x - y) \cdot (\nabla e_k(x) - \nabla e_k(y)) + d_r(e_k(x) + e_k(y)), \quad d_r = ce^{-r\ell_k}. \end{aligned}$$

The smooth rows vanish and its diagonal is $2d_r e_k(x)$. For $0 < r \leq r_0 \leq 1/2$,

$$\mathbb{E}|j_k - j_{k,r}| \leq C_k(r + (Nr)^{-1}), \quad \mathbb{E}|j_{k,r}|^2 \leq C_k(1 + \log(1/r) + (Nr^2)^{-1}).$$

The first estimate retains the correct heat response; the second uses the small-ball bound and layer-cake after truncation. No untruncated square estimate is used.

The Fourier coefficients of $j_{k,r}(x, y)\overline{e_k(x)}$ are

$$b_r(m) = c\{(k \cdot m)a_r(m) - (k \cdot (m + k))a_r(m + k)\} + d_r(\mathbf{1}_{m=0} + \mathbf{1}_{m=-k}).$$

They satisfy $b_r(0) = b_r(-k) = 0$, $\sum_{m \neq 0} b_r(m) = 2d_r$ and $|b_r(m)| \leq C_k a_{r/8}(m)$ at every nonzero frequency. Hence

$$\mathbb{E}[j_{k,r}(X_1, X_2)\overline{e_k(X_1)}] = \frac{N}{N-1} \mathbb{E} \sum_{m \neq 0} b_r(m) |Z_m|^2 - \frac{2d_r}{N-1}.$$

Set $A_2(t) = \mathbb{E}[j_k(X_1(t), X_2(t))\overline{e_k(X_1(t))}]$ and $B_2(t) = \mathbb{E}[j_k(X_1(t), X_2(t))\overline{e_k(X_1(0))}]$. On every fixed window $0 < \lambda_- \leq \beta/\sqrt{N} \leq \lambda_+ < \infty$, all $N \geq 2$ obey

$$\sup_{t \leq T} |A_2(t)| \leq CN^{-1/2}, \quad \sup_{t \leq T} |B_2(t)| \leq CN^{-1/4} \sqrt{1 + \log N}.$$

Indeed choose r a fixed multiple of $N^{-1/2}$. The complete drift-square energy identity gives tagged mean-square displacement $O(N^{-1/2} + \nu)$; Cauchy–Schwarz is applied only to the truncated kernel and displacement.

The remaining assertion. Define A_3, B_3 by replacing only the last label above by 3, and retain the same current or initial time. With $\tilde{a} = a - c/N$, the R25 signed remainder differs by at most $CN^{-1/4}\sqrt{1 + \log N}$ from

$$\mathcal{R}_N^{(3)} = \frac{(N-1)(N-2)}{2N} \int_0^T \left[e^{-2\tilde{a}(T-t)} A_3(t) - e^{-aT} e^{-\tilde{a}(T-t)} B_3(t) \right] dt,$$

interpreted as zero at $N = 2$. Thus its zero-limit and positive-limsup alternatives agree with those for the original signed remainder, for the same fixed data. The three-label term has not been shown to vanish and no admitted witness is known. The critical regime $\beta_N/\sqrt{N} \rightarrow \lambda > 0$, microscopic subcriticality $\beta_N/\sqrt{N} \rightarrow 0$, and the old energy-floor condition $\beta_N \rightarrow 0$ remain distinct.